
SMART COMPLAINT MANAGEMENT SYSTEM

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ABSTRACT

The Smart Complaint Management System is a web-based platform designed to streamline the process of lodging and addressing grievances in educational institutions. Traditionally, complaint registration was handled manually through paper-based systems, which were time-consuming, inefficient, and lacked transparency. This system provides a user-friendly and secure interface for students, faculty, and staff to submit complaints online, track their status, and receive timely resolutions. The system ensures confidentiality by allowing anonymous submissions, thus encouraging users to report issues without fear of backlash. Complaints are categorized and automatically assigned to the respective administrative departments for efficient handling. The admin dashboard provides a centralized control panel where grievances can be monitored, updated, and resolved in a structured manner. Key features include user authentication, role based access (Admin/User), complaint tracking, status updates, and report generation for data driven decision-making. The system is developed using D-Jango for backend processing, MySQL for database management, and Python for Frontend Processing, ensuring scalability and ease of use. By automating the grievance redressal process, this system enhances transparency, accountability, and efficiency, ultimately fostering a more responsive and supportive environment for students and staff.

Keywords: Complaint Management, Grievance Redressal, Web-Based Application, User Authentication, Role-Based Access.

I. INTRODUCTION

The Smart Complaint Management System is an online platform of private or public institutions enabling prompt actions on any issue raised by them. The cell members deal with all types of grievances, received from students. The implementation of the grievance system will be helpful to address all the grievances in a time-bound manner. This system helps to take quicker action on the grievance raised. The "Smart Complaint Management System" is a pioneering digital platform designed to streamline and enhance the grievance handling process in educational institutions. This web-based system provides a confidential, efficient, and user-friendly medium for students to report and resolve their grievances, fostering a safer and more supportive environment. user-friendly medium for students to report and resolve their grievances, fostering a safer and more supportive environment. The primary aim of this system is to establish a transparent and responsive channel for addressing various student grievances, ranging from academic issues to personal and infrastructural concerns. By digitizing the grievance process, the system ensures that students' concerns are heard, tracked, and addressed in a timely and effective manner. We are developing an online complaint Management System for submitting complaints online. By using complaint Management System a user can upload/post his complaint from anywhere by using this website on his phone or PC Online. Complaint Box is an easy and secure way as compared to other previous methods. Due to an online complaint box manual effort and wastage of paper is decreased. User can submit his complaint by easily creating his/her own profile also User can check the current status of their complaints and can view what kind of action is taken of his/her complaint. Online complaint Management System shows the current status of the complaint whether it is in process or closed. It is based on centralized management only the admin can check or solve the complaint. Admin having the authority to remove a User. Centralized management for checking current status of complaints and updating status of complaints. Admin can generate a report of this system in between selected dates of his own choice.

II. METHODOLOGY

The development of the Smart Complaint Management System followed a structured and phased approach. An initial system analysis was conducted to study the drawbacks of the existing manual grievance system. Key requirements such as transparency, accessibility, and tracking were identified. A modular design was adopted to separate functionalities for students, staff, and admin. Django framework was used for backend development to ensure scalability and rapid deployment. Python was chosen for logic implementation and MySQL for robust database management. User interface was designed using HTML and CSS for ease of navigation and accessibility. Input forms were designed to be intuitive and error-free for efficient data entry. Role-based access was enforced to control system permissions and maintain security. Complaint categories and workflows were automated to streamline resolution. Real-time status tracking and notifications were built in to improve user engagement. Admin module included features for complaint allocation and report generation. System design was supported with Data Flow Diagrams (DFDs) and flowcharts. Testing was carried out in phases: unit testing, integration testing, and system testing. User Acceptance Testing (UAT) involved feedback from students and staff users. Issues identified during testing were resolved to ensure a smooth user experience. The system was deployed on a web server for easy access from any device. Help documentation and training demos were prepared for user onboarding. Maintenance planning was done to ensure long-term usability and updates. The methodology ensured a secure, responsive, and effective complaint management system.

III. SYSTEM DESIGN

The Complaint Management System plays a crucial role in maintaining a responsive and efficient environment. Proper handling of complaints can significantly impact user satisfaction and trust, making it essential for organizations to establish a systematic and transparent process.

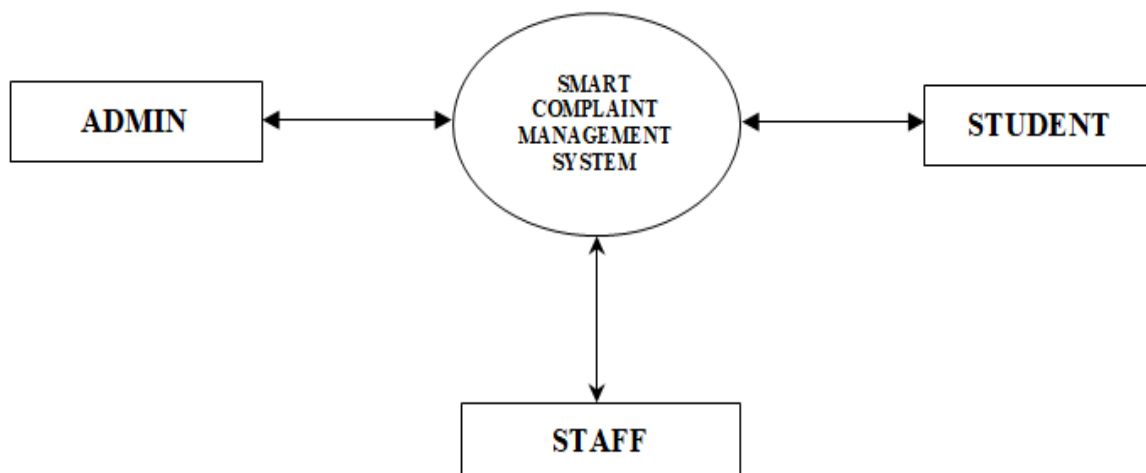


Figure 1: Data Flow

The Smart Complaint Management System is designed to digitize and streamline this process. The key steps involved in handling complaints are as follows:

1. Complaint Submission:

Users log into the system and register their complaints through an intuitive online form, optionally attaching evidence or selecting complaint categories.

2. Acknowledgment:

The system sends an automated acknowledgment to confirm that the complaint has been successfully received.

3. Verification and Assignment:

The admin or concerned authority reviews the complaint details and assigns it to the appropriate department or personnel for resolution.

4. Investigation and Resolution:

The assigned personnel investigate the issue and take necessary action to resolve it within a specified timeframe.

5. Status Tracking:

Users can monitor the progress of their complaint in real time, enhancing transparency and trust.

6. Closure and Feedback:

Once resolved, the complaint is marked as closed. Users may be prompted to provide feedback on the handling and outcome of the issue.

7. Reporting and Analysis:

Admins can generate reports to analyze trends, frequency, and performance in complaint handling for future improvements.

IV. IMPLEMENTATION

Module Description

1. Student Registration
2. Staff Registration & Login
3. Post Complaint
4. Process Module
5. Complaint Status
6. Pending Complaint
7. Allocation of Complaints to Respective Staff
8. Staff login
8. Reports

Database Design

This database used in this project is similar in structure to that of the relational database, while considering the database architecture there are two main categories such as relational database and document database.

In the document database the data values are maintained as a file in which all the information are stored in document. But the relational database is like tables, which are represented, in physical database as stored files. They have their own independent existence. A table consists of rows and columns. Each column corresponds to a piece of information called fields. A set of fields constitutes a record. The record contains all the information, specific to the particular item.

System Development

The key to controlling maintenance costs is to design systems that are easy to change, so the link between development and maintenance is very strong. Many of the analysis and design methodologies, tools, and techniques employed during system development can be applied to system maintenance, but there are significant differences between development and maintenance. Maintainability is the ease with which software can be understood, corrected, adopted and enhanced.

V. RESULTS

The Smart Complaint Management System was successfully developed and implemented as a web-based application to streamline grievance handling in educational institutions. The system enabled users to register, log in, submit complaints, and track their status in real-time. Role-based access ensured secure and organized handling of complaints by staff and administrators. Testing confirmed the effectiveness of each module, including automated acknowledgments, complaint categorization, and status updates. Admins were able to allocate complaints efficiently and generate detailed reports.

The system's performance was stable, with accurate data retrieval and storage through the MySQL database. Users found the interface easy to navigate, and the anonymity feature encouraged more open reporting. Overall, the system significantly improved transparency, reduced manual workload, and provided faster resolutions compared to traditional methods.

VI. OUTPUT

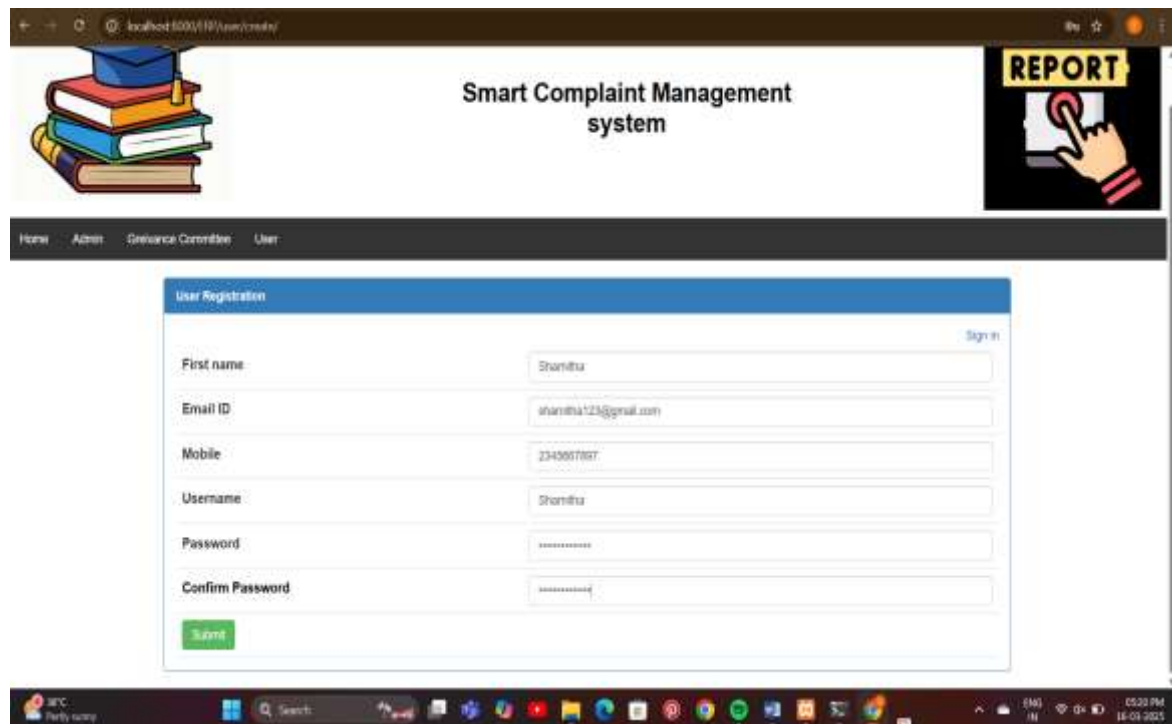


Figure 2: Register page

The registration page allows new users to create an account by providing essential details such as name, email, and password. It ensures secure access to the system and enables personalized complaint tracking.

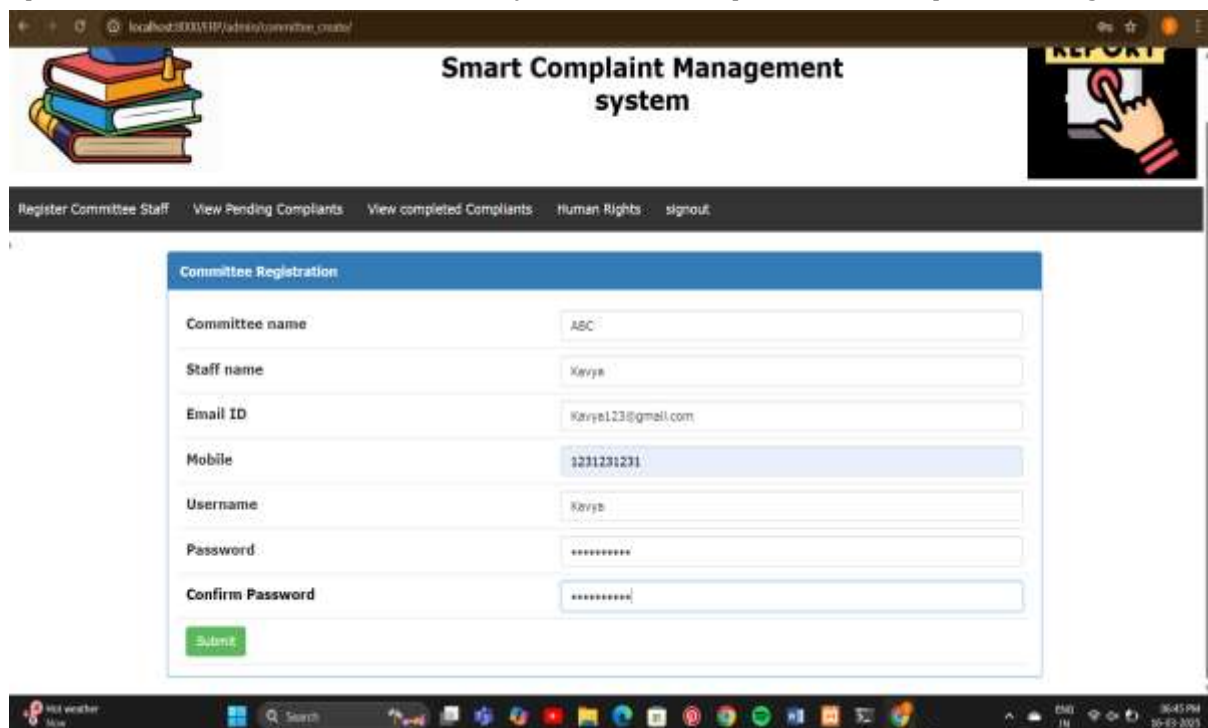


Figure 3: Staff registration page

The staff registration page is designed for administrative personnel to register into the system with designated roles. It enables secure access to manage, view, and respond to complaints assigned to their department.

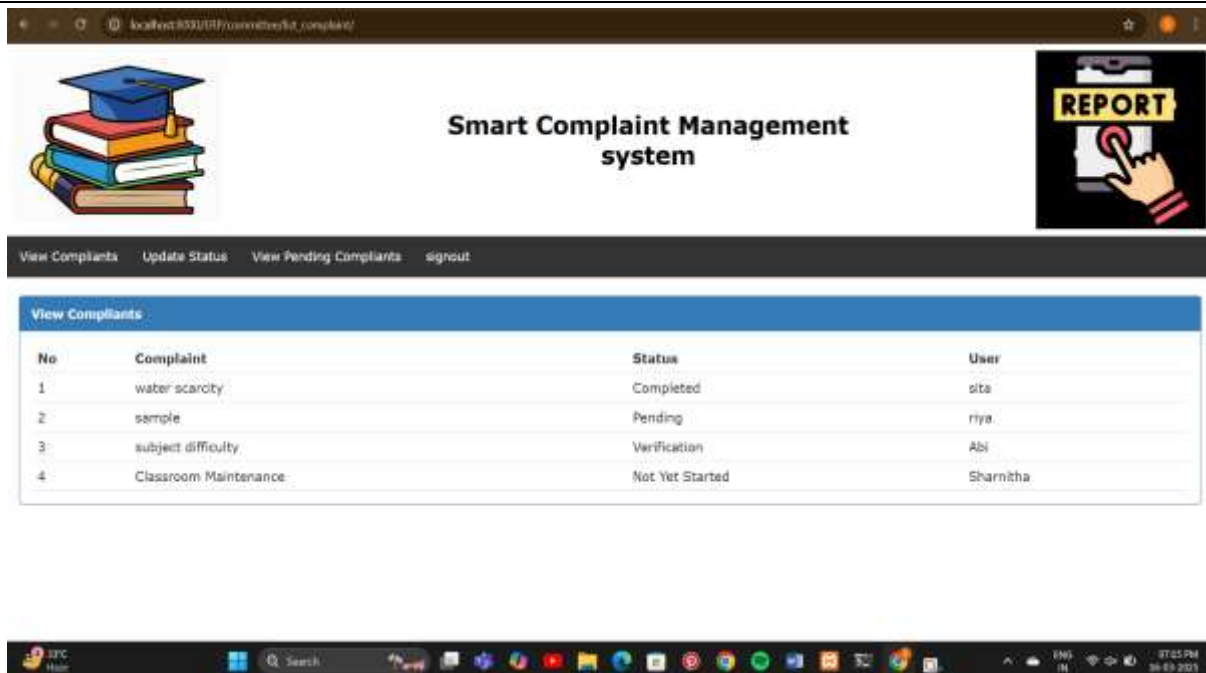


Figure 4: View complaint page

The View Complaint page allows users and staff to see the details, status, and history of submitted complaints.

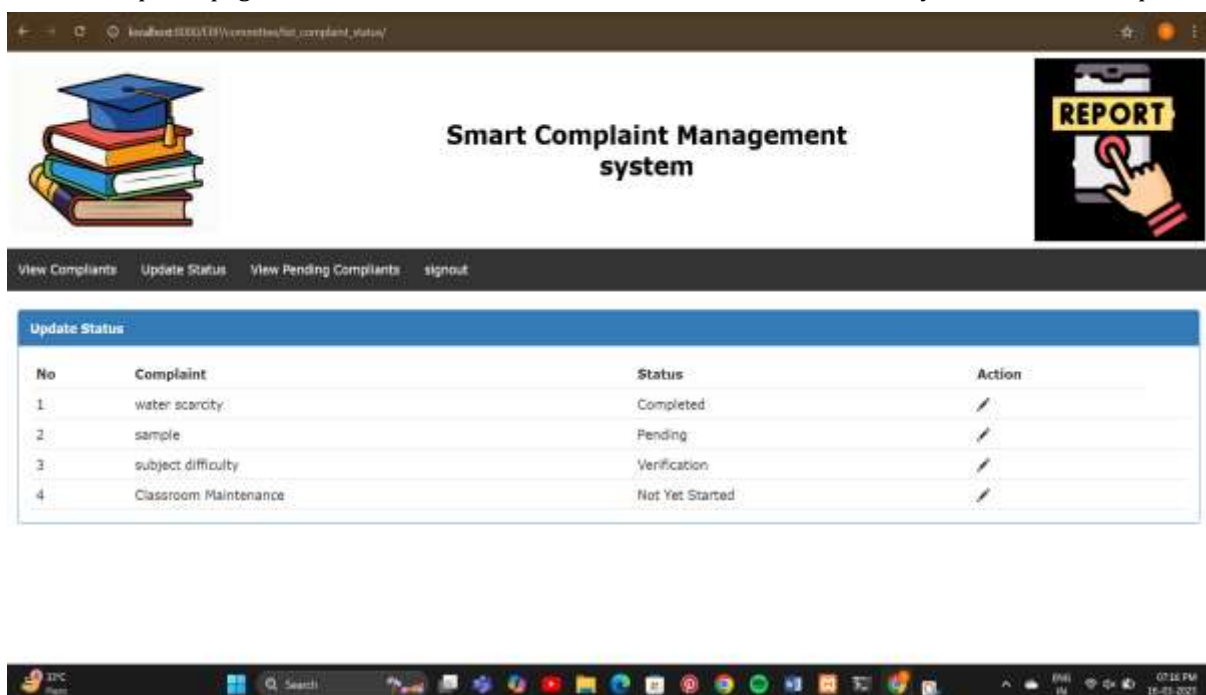


Figure 5: Status updation page

The Status Updation page enables staff to update the progress or resolution status of a complaint. It ensures users are informed in real-time about the handling of their issues.

VII. CONCLUSION

The computerized and online process significantly reduces common issues and saves time, allowing transactions to be viewed or retrieved at any level with ease. Error messages are provided at each stage of input to guide users and prevent mistakes. The software supports on-demand access to reports, making it easy for users to refer to necessary information whenever needed. Its menu-driven design, combined with simplicity and user-friendliness, makes the system highly accessible. Additionally, the software allows for easy

modification and maintenance, supports the storage and retrieval of large volumes of data, and ensures high speed and accuracy in operations. Data is entered in a structured and formatted manner, and reports can be generated in various formats. Most importantly, the system eliminates the need for manual processing in the user department, streamlining operations and improving overall efficiency.

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